Ch. 7. Entropy and free energy

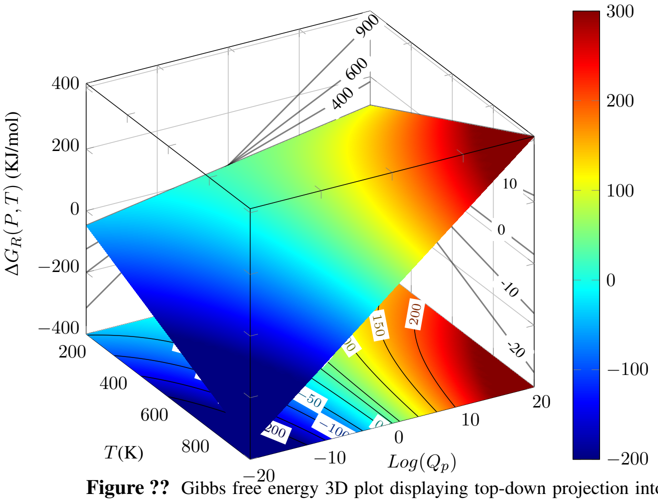

Figure ?? Gibbs free energy 3D plot displaying top-down projection into the LogQ_p-T bottom plane giving ΔG isolines, into the ΔG-T, right plane giving isobars with constant Q_p, and into the ΔG-LogQ_p, left plane giving isotherms with constant T. The color scale is blue for negative, small ΔG_R values, changing into red for positive, large ΔG values.

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7.1 Temperature and pressure effects on a reaction

Experimental evidence tells us that chemical reactions that produce heat or that produce entropy are likely to happen spontaneously. At the same time, chemical reactions with positive values of ΔG_R are not spontaneous. We can often adjust pressure or temperature, which we call the external conditions, to help spontaneity, hence changing ΔG_R from positive to negative. But those adjustments not always work. How can we promote spontaneity, and how can we rationalize the impact of entropy and entropy on spontaneity? This section addresses these two questions using Gibbs free energy 3D and 2D contour maps by examining the influence of external pressure and temperature conditions on spontaneity. These graphs, similar to weather maps used to visualize atmospheric data, are useful for getting a deeper understanding of chemical principles such as Le Chatelier's. The dependence of ΔG_R on temperature and pressure is given by:

$$\Delta G_R = \Delta H^\circ - T\Delta S^\circ + RT \cdot Ln10 \cdot \log Q_p$$

where we have used the properties of logarithms: $Lnx = Ln10 \cdot Logx$. ΔG_R is the force that drives a chemical reaction: the smaller and more negative this value, the more thermodynamically favoured the reaction will be.

`A two-dimensional function` ΔG_R data can be graphed either as a 3D graphs or as simplified-2D contour maps. On one hand, 3D ΔG_R plots are graphical 3D representations of the relationship between ΔG_R and its variables: p, and T (see inclined, colored plot in Figure **??**). On the other hand, 2D contour maps are two-dimensional visualizations representing ΔG_R data using lines of equal value called isolines or contour lines (see Figure **??** bottom, left, and right projections). 2D contour maps are generated by taking 3D ΔG_R graphs and projecting their constant height lines onto a 2D plane. These 2D plots use contour lines to represent 3D topographical data, where parallel slices at different values represent constant output values. For example, Figure **??** displays a top-bottom projection into the T-Q_p plane, a left-to-right projection into the ΔG_R-T plane, and a right-to-left projection into the ΔG_R-Q_p plane. 2D contour maps are widely used in scientific reporting, for example, in weather forecasting and meteorological analysis, to visualize complex weather patterns, helping simplify complex 3D visualizations.

`Isolines` As we can project $\Delta G_R(P,T)$ in three different planes, we can find three different types of isolines. First, if we project the 3D plot onto the LogQ_p-T plane, the bottom of the box in Figure **??**, we obtain isolines with constant ΔG_R value, called *isoenergy lines*. The equation for an isoenergy line is given by:

$$logQ_p = \frac{\Delta S^\circ}{R \cdot Ln10} + \frac{\text{const} - \Delta H^\circ}{R \cdot Ln10} \cdot \frac{1}{T}$$

These are curved lines with either a positive or a negative curve. A very important isoenergy line is the *equilibrium isoline* which corresponds to $\Delta G_R(P,T) = 0$ and has per equation

$$lnQ_p = \frac{\Delta S^\circ}{R \cdot Ln10} - \frac{\Delta H^\circ}{R \cdot Ln10} \cdot \frac{1}{T}$$

Second, if we project onto the ΔG-T plane, the right side of the box in Figure **??**, we obtain isolines with constant LogQ_p value called *isobars* as Q_p depends on the pressure of reactants and products. The equation for an isobar line is given by:

$$\Delta G = \Delta H^\circ - (\Delta S^\circ - R \cdot Ln10 \cdot \text{const})T$$

These are straight lines with either a positive or a negative slope. Third, if we project onto the ΔG-LogQ_p plane, the left side of the box in Figure **??**, we obtain isolines with constant T value called *isotherms*. The equation for an isotherm line is given by:

$$\Delta G = (\Delta H^\circ - \text{const} \cdot \Delta S^\circ) + (R \cdot \text{const} \cdot Ln10) lnQ_p$$

These are straight lines, always with a positive slope. These different types of isolines, their slope, and the sign of their curve, give important information about the energetics, the enthalpy, and the entropy of the reaction.

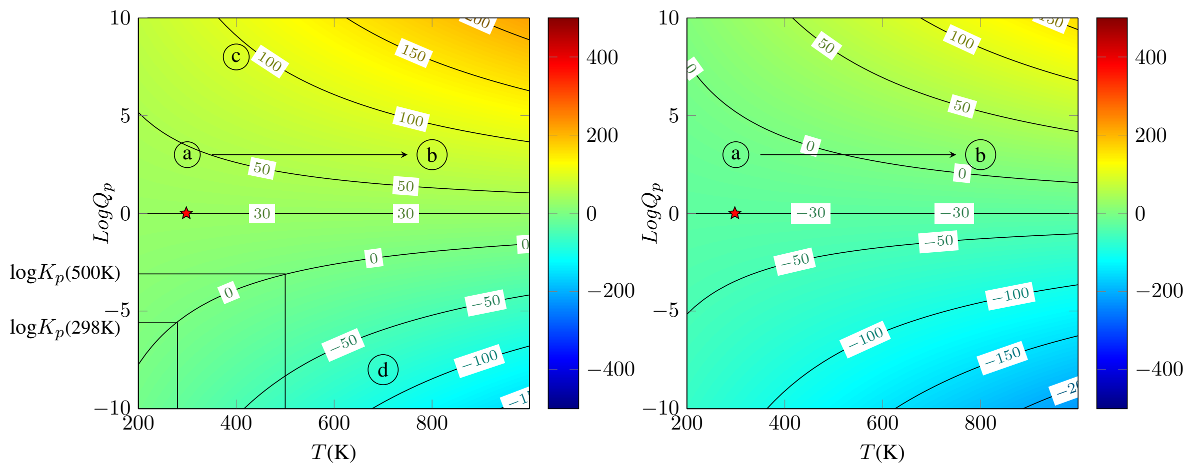


Figure ?? Gibbs free energy 2D plots displaying contour lines for different pressure and temperature conditions for an endothermic (Left Panel) and an exothermic (Right Panel) reaction. The star sign indicated the standard pressure and temperature conditions. The equilibrium isoline corresponds to $\Delta G_R = 0$.

`Interpretting contour maps` $\Delta G_R(P,T)$ contour maps, in particular those representing isoenergy lines, are useful graphical constructs. Information such as spontaneity, the thermal character (e.g., whether the reaction is endothermic or exothermic), the entropic character of a reaction (e.g., whether the reaction produces or consumes entropy) and even the equilibrium constant can be easily obtained just by inspecting the map. Perhaps teh easiest reaction characteristic that can be obtained from the $\Delta G_R(P,T)$ contour maps is spontaneity. The reaction is nonspontaneous under the temperature and pressure conditions represented in point *c* of Figure **??** (left panel), as $\Delta G_R(P,T)$ is positive. We say that the reaction is *endergonic*. Differently, under the temperature and pressure conditions represented in point *d* the reaction will proceed spontaneously as $\Delta G_R(P,T)$ is now negative and the reaction is now *exergonic*. The curvature of the equilibrium isoline informs about the thermal character of a reaction. The reaction depicted in Figure **??** (left panel) is endothermic, as the equilibrium isoline bends upwards, whereas the reaction depicted in Figure **??** (right panel) is exothermic, as the equilibrium isoline bends downwards. The Asymptotic values of the equilibrium isoline for large temperatures inform about the entropic character of a reaction. We can see in Figure **??** (left panel) that the equilibrium isoline tends to a positive value ($\sim$3) at very large temperatures, hence the reaction will produce entropy. Differently, the reaction depicted in Figure **??** (right panel) consumes entropy as the equilibrium isoline tends to a negative value ($- \sim$3) at very large temperatures. We can also ontain K_c for different tempeature using the equilibrium isoline. Remember that K_c changes with temperature and hence there is not a single value for K_c for a reaction. We can, for example, obtain K_c for standard temperature conditions (T=298K, the temperature at which all thermodynamic data is tabulated) as shown in Figure **??**. The equilibrium isoline either increases or decreases with temperature. As we described, the value of the isoline at fixed conditions informs about spontaneity, whereas the curvature of the equilibrium isoline informs about enthalpy, and the asymptotic values of the equilibrium isoline informs about entropy.

`Pressure effects` Now, let us address the effect of changing the pressure. Increasing the product's pressure, that is going from bottom to top of Figure **??**, will make $\Delta G_R(P,T)$ more positive, whereas increasing the reactant's pressure, that is going from top to bottom of Figure **??**, will make $\Delta G_R(P,T)$ more negative. This is a thermodynamic interpretation of the *Le Châtelier principle*: when we have many reactants and few products, the reaction will proceed to the right. Therefore, a very important idea to remember is that adding fewer products and more reactants than at equilibrium will *always* drive the reaction towards the right, helping spontaneity.

`Temperature effects` We may think that we can favour a reaction simply by increasing temperature. However, the dependence of Gibbs free energy on temperature is different for different LogQ_c values. For this reason, in fact, we do not always favour spontaneity just by increasing the temperature. Let analyze an experiment in which we increase the temperature going from points *a* to *b* following the arrow in Figure **??**. For example, in Figure **??** (Left panel), we have that the reaction is nonspontaneous in both *a* and *b* temperature conditions. The increasing of temperature did not help spontaneity. Differently, on Figure **??** (Right panel), we have that the reaction is not spontaneous in in *a* but becomes spontaneous in *b*. That is, increasing the temperature did promote spontaneity. Hence, selecting the right value for LogQ_c, that is the right ratio between reactants and products is important.

For very small LogQ_c values, the isoenergy lines increase with temperature, curving upwards. This means that when the temperature increases, ΔG_R becomes more negative, hence favouring spontaneity. In this situation, the thermodynamic force that drive the reaction would become stronger. For large LogQ_c values, the isolines decrease with temperature, curving downwards. This means that when the temperature increases, ΔG_R becomes less negative, hence disfavouring spontaneity. There is a crossing LogQ_c value (LogQ_c^*) that separates isolines curving upwards and downwards given by

$$\boxed{\text{Log}Q_c^* = \frac{\Delta S^\circ}{Ln10 \cdot R}} \tag{7.1}$$

In short, LogQ_c needs to be smaller than entropy so that temperature can promote a reaction. Only for LogQ_c values smaller than LogQ_c^*, we can favour spontaneity by increasing temperature. For example, on Figure **??** (Left panel) LogQ_c^* is approximately -3, whereas on the right panel, approximately +3. But, even when LogQ_c is smaller than the critical value, the temperature has to be large enough for a reaction to proceed spontaneously. For low temperature, even when we do not cross Log$Q_c(critical)$, the reaction will not proceed spontaneously. The crossing temperature value is given by

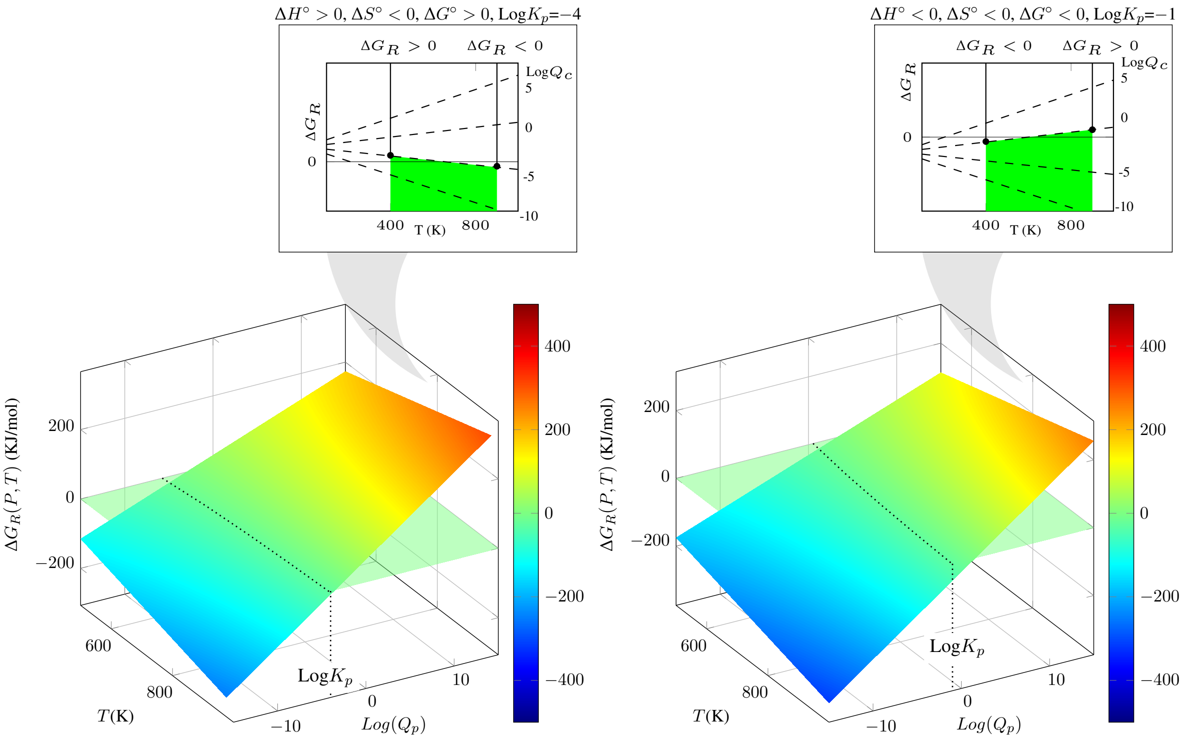
$$\boxed{T^* = \frac{\Delta H^\circ}{\Delta S^\circ - R \cdot Ln10 \cdot \text{Log}Q_c}} \tag{7.2}$$

The crossing values for T and Q_c are different by nature. By comparing Equations and we could see that while the crossing temperature value depends on ΔH°, ΔS° and LogQ_c, the crossing Q_c value only depends on ΔS°.

The information presented here demonstrates how a combination of low enough pressure conditions and large enough temperatures is needed for a reaction to happen spontaneously.

`Interpretting 3D maps` Exothermic reactions, think of an explosion, are more likely to happen spontaneously, whereas reactions producing entropy, think of a decomposition in which a large molecule splits in two, are also more likely to happen without help. Why do some reactions happen spontaneosuly whreas other do not? We will analyze a series of 3D Gibbs free energy maps in order to rationalize spontaneity. Figure **??** displays a series of 3D Gibbs free energy maps. We can see that these are not flat maps, and the maps twist upwards in one extreme and downwards in the other along the temperature axis. That explains why the isobars, obtained by flattening the maps in the T-ΔG_R plane shown in the inserts, have different slopes of increasing value from negative to positive. First, if we compare Figure **??** (Top, Left panel) with (Top, Right panel), we would see that the lower the enthalpy, the lower the map, as it implies a downward shifting of the maps. A similar effect is found when comparing Figure **??** (Bottom, Left panel) with (Bottom, Right panel). This explains why exothermic reactions are often more likely to be spontaneous, as small, negative enthalpies increase the force driving chemical reactions, ΔG_R. Similarly, if we compare Figure **??** (Top, Left panel) with (Bottom, Left panel), we would see that the larger the entropy, the lower the positive and the negative twist, which effectively shifts the ΔG_R graph down, hence making the plot more negative. A similar change is found when comparing Figure **??** (Top, Right panel) with (Bottom, Right panel). That is why reactions with entropy production are often more likely to be spontaneous, as large positive entropies increase ΔG_R. Overall, we have that decreasing enthalpy shifts ΔG_R down while increasing entropy twists the same plot downwards, both promoting the force driving chemical reactions.

Summary In summary, the pressure and temperature conditions are tangled in their dependence on $\Delta G_R(P,T)$, whereas only temperature affects $\Delta G_R^\circ(T)$. We can always adjust the pressure ratio to decrease $\Delta G_R(P,T)$, helping spontaneity. Pressure effects always follow the same trend: adding more products and fewer reactants than at equilibrium will always help spontaneity. Differently, the impact of temperature on equilibrium varies, depending on the sign of $\Delta G_R^\circ(T)$: for negative $\Delta G_R^\circ(T)$ values increasing temperature helps spontaneity, whereas for positive $\Delta G_R^\circ(T)$ values decreasing temperature helps spontaneity.



▲An endothermic reaction with entropy consumption ($\Delta H^\circ = +30\text{KJ}\cdot\text{mol}^{-1}$, $\Delta S^\circ = -50\text{J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$, Log$K_p$=−4)
▼An endothermic reaction with entropy production ($\Delta H^\circ = +30\text{KJ}\cdot\text{mol}^{-1}$, $\Delta S^\circ = +50\text{J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$, Log$K_p$=+1)

▲An exothermic reaction with entropy consumption ($\Delta H^\circ = -30\text{KJ}\cdot\text{mol}^{-1}$, $\Delta S^\circ = -50\text{J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$, Log$K_p$=−1)
▼An exothermic reaction with entropy production ($\Delta H^\circ = +30\text{KJ}\cdot\text{mol}^{-1}$, $\Delta S^\circ = +50\text{J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$, Log$K_p$=+4)

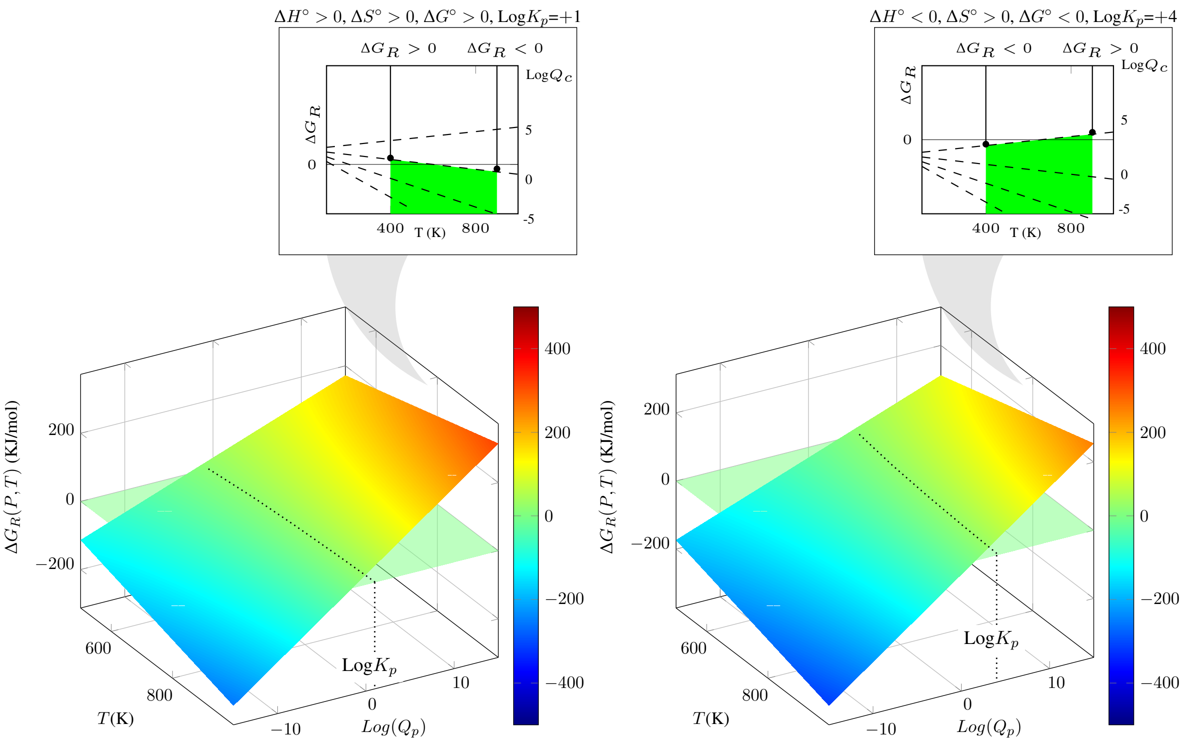


Figure ?? Gibbs free energy 3D plot for a set of reaction with different internal thermodynamics parameters: (Top Left) an endergonic reaction: you can increase T to favour the reaction only for pressure conditions (LogQ$_c$) lower than $\Delta S/R$); (Top Right) an exergonic reaction mediated by H: you disfavour the reaction by increasing T only for pressure conditions (LogQ$_c$) higher than $\Delta S/R$); (Bottom Left) an endergonic reaction mediated by S: you can increase T to favour the reaction only for pressure conditions (LogQ$_c$) lower than $\Delta S/R$); (Bottom Right) an exergonic reaction mediated by both the H and S: you disfavour the reaction by increasing T only for pressure conditions (LogQ$_c$) higher than $\Delta S/R$);